Contents lists available at ScienceDirect

Computers & Graphics

journal homepage: www.elsevier.com/locate/cag

Modeling luminance contrast for optical see-through displays

ARTICLE INFO

Article history:
Received August 24, 2022

2010 MSC: 68U05

Keywords: Augmented Reality, Optical See-Through Displays, luminance contrast, visual perception

ABSTRACT

Due to the *additive light model* employed by current optical see-through head-mounted displays (OST-HMDs), the perceived contrast of displayed imagery is reduced with increased environment luminance, often to the point where it becomes difficult for the user to quickly and accurately distinguish visual features within displayed virtual imagery. This problem is particularly important when it comes to user interface (UI) design, where the effectiveness of the UI is impacted both by the colors of the UI, the particular display being used, and the characteristics of the observer's physical environment. While existing contrast models can be used to predict user performance involving visual tasks on more traditional displays, they must be adapted for use with additive displays in order to take into account the blending of environment light with light emitted by the display. In this paper, we present a simplified model of luminance contrast for optical see-through displays that is derived from Michelson's contrast equation, and validate its performance by comparing predicted contrasts to contrasts calculated based on a set of photometric measurements made on a consumer OST-HMD, the Microsoft HoloLens 2. We demonstrate how this model can be used to inform UI design decisions by helping to select colors that are more robust to the observer's physical environment conditions, and demonstrate several additional use cases.

1. Introduction

Optical see-through head-mounted displays (OST-HMDs) allow users to view virtual imagery that is superimposed over their view of their physical surroundings. These devices use what is commonly known as the *additive light model*, since the light emitted by the display is added with the light originating from within the user's environment. These particular displays are prone to unique perceptual issues compared to other more traditional displays, in that the perceived appearance of virtual imagery tends to differ from its intended appearance as the user observes a blending between light emitted by the display and light from within their physical environment. This blending of light leads to imagery tending to appear transparent [1, 2, 3]. Additionally, these displays typically cannot render colors that appear darker than the physical environment in which they are being presented, as to achieve this, light would need to be attenuated from the user's view of their environment [4, 5]. To use photometric terminology, colors with less luminance than the environment luminance from the observer's perspective cannot be displayed by the optical see-through display. While these effects are not problematic when observing imagery in an environment with low luminance, such as in a dark room, the perceived range of colors that the display is able to produce shrinks with increased environment luminance, since more colors fall below this luminance threshold set by the environment conditions. As a result of this, the perceived contrast of virtual imagery shown on the display similarly decreases with increased environment luminance, as there is a less apparent difference in lighting between different points within the virtual imagery [6].

Poor contrast leads to usability and perceptual issues where the observer may be unable to accurately distinguish visual features within the virtual imagery (e.g., reading text on the display), and in extreme cases, such as sunny outdoor lighting conditions, the user may not even be able to recognize the presence of the virtual image at all [7, 8, 9]. By identifying when problematic conditions occur, we set the stage for potential methods that improve image quality, whether these methods rely on high-level software-based contrast improvement strategies, or more fundamental changes to the underlying design of the display and optics. For this reason, it is important to be able to recognize when environment conditions are likely to interfere with the user's ability to perceive virtual imagery shown on optical see-through displays.

While there are several commonly used contrast models, such as Michelson contrast [10], Weber contrast [11], and root mean square (RMS) contrast [12], these models must be adapted for use with optical see-through displays in order to create separate input terms for describing the light emitted by the display and the lighting conditions within the observer's environment. To our best knowledge, such a model does not exist, and as the main contribution for this paper, we demonstrate how the Michelson contrast equation can be adapted for better use with optical see-through displays. Our adapted model predicts luminance contrast given input in the form of an image to be displayed, a set of parameters that describe the optical see-through display, and an approximation of the observer's physical environment conditions. We validate the performance of this model by comparing its predicted contrasts against contrasts calculated from direct photometric measurements on one commonly used optical see-through display, the Microsoft HoloLens 2, and show that our model achieves a high degree of accuracy with a residual standard error of $RSE = 3.2\%$ and $R^2 = 98.0\%$.

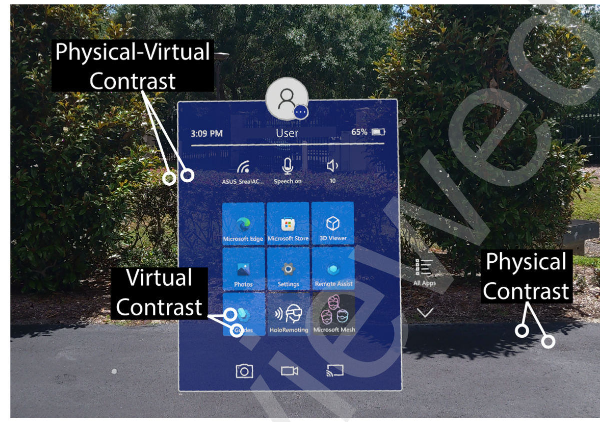


Fig. 1. Three types of contrast comparisons can be made from the observer's perspective within an OST AR display. Physical-virtual contrast compares between a point in the AR imagery and a point in the observer's physical environment, in this case between the floating panel and the bushes behind it. Virtual contrast compares between two points within the AR imagery, in this case between the white text and blue background. Finally, physical contrast compares between two points in the observer's physical background.

We describe the potential use cases of this model, including how it can be used to identify displayed color pairings that are prone to be perceived with low contrast for a given environment condition and set of display parameters. We additionally show how the model can be used to find the maximum range of environment conditions in which a particular display configuration can be used, as well as show how this range can be extended by increasing the attenuation factor of the display, such as by incorporating an additional neutral density filter. Finally, we offer guidelines for user interface (UI) design on OST-HMDs based on the understanding provided by this model.

The remainder of this paper is structured as follows: Section 2 presents background information on the terminology used in this paper and the factors that influence the luminance contrast of augmented reality (AR) imagery on OST-HMDs. Section 3 presents the equations defining our model of luminance contrast for OST displays. Section 4 outlines the methods used to make photometric measurements on the HoloLens 2 and creates a data set that can be used to evaluate the performance of the model. Section 5 describes an analysis of the performance of the contrast model. Section 6 provides a general discussion about the applications of the contrast model. Section 7 concludes the paper.

2. Background

In this section, we provide a brief overview of the terminology used in throughout the paper and introduce related work involving luminance contrast and AR.

2.1. Terminology

Throughout this paper, we frequently refer to photometric measurements of *illuminance* and *luminance* [13, 14]. Illuminance is a measure of the perceived intensity of light incident along a given surface. Illuminance values have units of *lux* (lx), lumen per square meter (lm/m^2), or candela-steridian per square meter ($cd \cdot sr/m^2$). Illuminance is different from *luminance*, in that luminance measures the perceived intensity of light traveling in a particular direction, and has units of candela per square meter (cd/m^2), or lumen per steridian square meter ($lm/(sr \cdot m^2)$). While *luminance contrast* is typically calculated by comparing the ratio of luminance values, since luminance and illuminance differ only by a scale factor represented by the steridian term, luminance contrast can be calculated through either illuminance measurements or luminance measurements. Throughout this paper, the luminance contrast values that we calculate are based on illuminance measurements.

While there are several methods of calculating luminance contrast, we will focus on the equation proposed by Michelson, which is commonly used when there is not necessarily a uniform background for which visual features are presented upon [10]. This is a common occurrence for OST AR, as the user's view of their physical environment provides a typically non-uniform background for which virtual imagery is superimposed over.

$$C = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} \tag{1}$$

Here, I_{max} is the greater of the two illuminance values, and I_{min} is the lesser.

In terms of human visual perception, higher contrast is generally considered better when it comes to making a visual stimulus more easily distinguishable, e.g., when designing AR user interfaces or presenting text [15]. However, it has been suggested that the range of usable luminance contrasts may have upper limits beyond which greater contrast may result in decreased legibility of stimuli [16], although the reasons for why this occurs are not completely understood.

2.2. Contrast of AR Imagery

With OST AR imagery, there are several ways in which luminance contrast can be compared. It can be compared between two points in the user's physical environment seen through the display (physical contrast), between two points in the virtual image itself (virtual contrast), or between a point in the user's physical environment and a point in the virtual image (physical-virtual contrast). Since black imagery is completely transparent on OST displays, physical-virtual contrast is essentially a special case of virtual contrast, where one of the points of interest is black/transparent in the virtual image. This point could be a point within virtual imagery that was intended to appear black, or a point within the field of view of the display in which no virtual imagery is being presented. The differences between these three contrast comparisons can be seen in figure 1. For the work presented here, we focus primarily on virtual contrast and physical-virtual contrast.

When it comes to AR virtual imagery on an OST-HMD, there are several factors that can lead to poor luminance contrast, which may make it difficult for the user to identify visual features within the image. These factors include environmental factors such as the intensity of environment lighting conditions and the coloration of things within the user's environment, as well as display factors, such as the maximum luminance of the display, the attenuation employed by the display, and the colors that compose the virtual image. Tinted visors are often integrated into the design of OST displays as a means for improving luminance contrast, where attenuating environment lighting as it combines with the display-emitted light improves virtual contrast and physical-virtual contrast. However, increasing the amount of attenuation inherent to the display reduces physical contrast, which changes the manner in which users perceive color in their environment, and may lead to issues with navigating dimly-lit environments [17, 18].

To understand the effects of low luminance contrast on

HMDs, Harding et al. [19] presented a simulation of different see-through backgrounds with overlaid white-color symbols. They found a strong effect of the visual complexity of the physical backgrounds on participants' perception of the virtual symbols. Similar to OST-HMDs, Merenda et al. [20] investigated in-vehicle heads-up displays (HUDs), where the virtual imagery blends with the appearance of the physical environment (e.g., road) behind it. In a user study, they found that the luminance and chromaticity of the virtual imagery both had effects on participants' choices. Participants generally chose colors with higher luminance, and some colors, in particular blue, green, and yellow could be more easily identified by their participants.

Gabbard et al. [1] discussed the effects of physical-virtual contrast between text and physical backgrounds on OST AR displays, highlighting effects of and interactions between the visual complexity of the physical background and the drawing style of the virtual text. Most techniques aimed at improving the legibility of virtual text either alter the appearance of the text itself, alter the appearance of the area around the text, or reposition the text to a more suitable location. For instance, Gabbard et al. [21], Kim et al. [22], and Erickson et al. [23] found that presenting light text over dark backgrounds on OST-HMDs generally results in increased legibility and user experience compared to dark text over light backgrounds, which differs from results for other displays like virtual reality HMDs [24]. Jia et al. [25] further found that virtual text presented over uniform surfaces is preferred by participants, leading them to develop a method for the automatic placement of AR textual annotations over appropriate surfaces using computer vision. Orlosky et al. [26] further presented a method for the automatic placement of AR annotations that prioritizes locations that are uniform and dark.

Other work focused on AR display techniques and devices that support luminance contrast and faithful color reproduction. For instance, Itoh et al. [27] presented a method to calibrate colors on OST-HMDs by processing the virtual imagery on the display to match the perceived colors by the users. They further presented a novel light attenuation method based on spatial color subtraction instead of an additive light model [4]. Similarly, Hincapié-Ramos et al. [28] introduced methods for processing the appearance of text to preserve intended colors while also increasing contrast. Leykin and Tuceryan [29] leveraged machine learning to predict the legibility of virtual text. Donval et al. [30] further presented a smart filter for OST-HMDs that could adapt the filter transmission for different background illumination conditions.

In 2021, Zhang and Murdoch demonstrated that the contrast between a virtual image on an OST display and the physical environment behind it can be used to predict how transparent users will perceive the imagery to be [31]. They present a model that accurately predicts perceived transparency as a function of contrast, which is important to consider when designing applications for OST displays since perceived transparency has been shown to have other non-intuitive effects, such as affecting perceived humanness of virtual humans, and affecting perception of ground contact [32, 33].

3. Luminance Contrast Model

In this section, we derive a luminance contrast model for OST displays that predicts contrast between two colors within the display's color space for a given set of parameters associated with the device's capabilities and a parameter associated with the lighting conditions of the observer's physical environment.

As mentioned in section 2, luminance contrast can be calculated using terms of either illuminance or luminance (but not a mix of the two). Illuminance values can be measured with a common light meter, whereas luminance values require more specialized and expensive equipment. Therefore, for the sake of accessibility, we choose to use illuminance values for the terms in our equations presented in this section. Note that in practice, either is acceptable, and therefore the equations can be used with either illuminance or luminance terms.

The luminance contrast of the imagery shown on the display is dependent on the lighting conditions in the user's environment, and Michelson's equation for luminance contrast (equation 1) can be manipulated to reflect this. In this case, the maximum and minimum illuminance terms in the equation can be considered as the sum of light originating from within the observer's environment with the light being emitted by the display. Here, the terms I_a and I_b represent the illuminance from the observer's eye position within the OST display for a chosen color a and b respectively, without regard to environment lighting conditions (i.e., in a pitch dark room). Whereas, I_{env} is the illuminance from the observer's eye position corresponding solely to the light coming from their physical environment (i.e., with the display turned off).

$$I_{max} = max(I_a, I_b) + I_{env} \tag{2}$$

$$I_{min} = min(I_a, I_b) + I_{env} \tag{3}$$

By substituting equations (2) and (3) into equation (1) and simplifying, we obtain the following version of the Michelson contrast equation, that considers the user's environment lighting conditions:

$$C = \frac{max(I_a, I_b) - min(I_a, I_b)}{I_a + I_b + 2I_{env}} \tag{4}$$

This expression can be simplified by introducing absolute values to eliminate the min and max expressions. As mentioned previously in section 2, many OST displays incorporate a tinted visor to attenuate light from the user's environment as it blends with light emitted by the display. To consider this attenuation of environment light, we introduce an attenuation factor, $k \in \mathbb{R} : 0 \leq k \leq 1$, which scales I_{env}:

$$C = \frac{|I_a - I_b|}{I_a + I_b + 2kI_{env}} \tag{5}$$

This equation forms the basis for our model of luminance contrast for OST displays, however this equation requires knowing I_a and I_b, the illuminance contributed by two arbitrary colors, a and b, shown on the display. Since these values are not usually known, we need to establish a function that outputs the illuminance contributed for a color input by its RGB value, where $r, g, b \in \mathbb{R} : 0 \leq r, g, b \leq 1$. The illuminance of any color shown on the display is the sum of the illuminance occurring from each of the primary color channels of the display, which contribute maximum illuminance values of I_r, I_g, and I_b for the red, green, and blue color channels respectively. Therefore, it is intuitive to think that the function would be a linear combination of maximum illuminance values of the primary color channels of the display and RGB values of the displayed color in the following form:

$$I(r, g, b) = I_r r + I_g g + I_b b \tag{6}$$

However, in practice, gamma correction is typically applied in the form of a transfer function so that step-wise changes in RGB values occur in perceptually equal intervals of lightness [34]. This correction is typically in the form of a power function we will call G, where $\mathrm{G}(x) = ((x+c_1)/c_2)^\gamma$. The c_1, c_2, and γ values are defined according to the color and image encoding standards being used by the specific display, such as standard red, green, blue (sRGB) or the BT.709 standard [35, 36]. Therefore, to factor gamma correction into the equation, we pass r, g, and b through the function G:

$$I(r, g, b) = I_r \mathrm{G}(r) + I_g \mathrm{G}(g) + I_b \mathrm{G}(b) \tag{7}$$

This equation can be simplified using vector notation, as a scalar product between two vectors, where the maximum illuminance of the primary color channels are expressed as a row vector $\mathbf{a}$, where $\mathbf{a} = [I_r, I_g, I_b]$, and the gamma corrected RGB values are specified as a row vector $\mathbf{x}$, where $\mathbf{x} = [\mathrm{G}(r), \mathrm{G}(g), \mathrm{G}(b)] = [r', g', b']$. Here, ^{T} is used to represent the transpose of the vector it is applied to.

$$I(\mathbf{x}) = \mathbf{a}\mathbf{x}^T \tag{8}$$

This equation can be substituted back into equation (5) to yield the main equation of our new model of luminance contrast for OST displays:

$$C = \frac{|\mathbf{a}(\mathbf{x} - \mathbf{y})^T|}{\mathbf{a}(\mathbf{x} + \mathbf{y})^T + 2kI_{env}} \tag{9}$$

The different parameters of the equation are defined as follows:

- $\mathbf{a} = [I_r, I_g, I_b]$: A row vector representing the illuminance from the observer's eye position of the red, green, and blue color channels respectively, at maximum intensity and in a pitch dark environment;

- $\mathbf{x}, \mathbf{y} = [G(r), G(g), G(b)] = [r', g', b']$: Row vectors containing the gamma corrected RGB values, r', g', and b', of the two colors being compared;
- k: The attenuation factor of the display;
- I_{env}: The illuminance from the observer's eye position contributed by the lighting in the observer's physical environment.

The final form of the equation can be used to predict luminance contrast values between any pair of RGB color coordinates in the device's color space. As we will demonstrate in the next section, several illuminance measurements can be made on the OST-HMD in a dark environment to find **a**, and within a lit environment to find I_{env} and k. Alternatively, I_{env} can be looked up from a table of common environment lighting conditions[1].

In order to evaluate the performance of this model, we need a set of luminance contrasts calculated based on photometric measurements made on an OST-HMD under varying lighting conditions and for different colors. In the next section, we present the methods behind our photometric measurements and present illuminance and contrast measures for one particular OST display, the HoloLens 2.

4. Photometric Measurements

This section describes the controlled photometric measurements we made with a current OST-HMD—a Microsoft HoloLens 2. These measurements demonstrate how to find the terms **a**, I_{env}, and k as described in the previous section, and provide the basis for evaluating the performance of our luminance contrast model.

4.1. Apparatus

We measured illuminance values on the HoloLens 2, which was released in 2019, and features a diagonal field of view of 52 degrees and a resolution of 2048×1080 pixels per eye. All photometric measurements were taken with the HMD powered on and with the brightness control settings of the device turned to 100%. Smaller percentages of display brightness were not tested, since we are primarily concerned with the maximum contrast able to be achieved under each tested lighting condition.

The illuminance measurements were made using an Urceri MT-912 light meter, which is reported to have an accuracy of ±3% of the measured value, and can make measurements between one and 200,000 lx, as reported by the Urceri website[2].

We aimed for illuminance measurements distributed across a wide range of common environment lighting conditions from dim indoor or nighttime lighting levels to bright sunny outdoor lighting levels. To do this, we chose to use a single dimmable Amzcool natural light lamp as our light source, since it could be easily adjusted to create illuminance measurements between 0 to 10,000 lx depending on the distance between the light meter and the light source. The environment in which the measurements were made had a consistent ambient illuminance of less than 1 lx, with no light sources other than the dimmable natural light lamp to influence the measurements. Environment lighting conditions with the lamp powered off were verified to be less than 1 lx.

When performing direct illuminance measurements of the light source, the light meter was positioned to be 8 cm away from the light source and parallel to its surface (see figure 2). This distance was chosen because direct illuminance measures taken of the light source at maximum intensity from this distance corresponded to our desired upper limit of 10,000 lx. When taking illuminance measurements through the display, the display was positioned so that the light meter was at the user's left eye position relative to the display. Relative to the light source, the light meter was in the same position as it was when making direct illuminance measurements of the light source (8 cm away and centered).

For virtual content presentation on the HoloLens 2, we used the Unity game engine. Imagery was streamed from a nearby laptop to the device via the holographic remoting feature. The laptop was positioned away from the measurement area to

[1] https://www.engineeringtoolbox.com/light-level-rooms-d_708.html

[2] https://www.urceri.com/mt-912-light-meter.html

Fig. 2. Illustration of the setup used to perform photometric measurements.

not affect the measurements. For each measurement, a single color was presented filling the entire field of view of the display. Colors were varied between 14 different RGB values representative of the primary secondary and tertiary colors as well as black and white: black RGB(0,0,0), white RGB(1,1,1), red RGB(1,0,0), green RGB(0,1,0), blue RGB(0,0,1), yellow RGB(1,1,0), cyan RGB(0,1,1), magenta RGB(1,0,1), orange RGB(1,0.5,0), chartreuse RGB(0.5,1,0), spring green RGB(0,1,0.5), azure RGB(0,0.5,1), violet RGB(0.5,0,1), rose RGB(1,0,0.5). The primary, secondary, and tertiary colors were displayed at maximum saturation and value. The inclusion of the black and white conditions allow us to establish the maximum contrast for each environment lighting condition, while the colors sample even intervals within the display's color space for single color channels (primary color), and combinations of two color channels (secondary and tertiary colors).

4.2. Methods

The illuminance measurements were made under the following controlled lighting conditions:

Environment Lighting (5 levels): The dimmer on the environment light source was varied to achieve illuminance values of 10,000, 1,000, 500, 100, and 0 lx. These light levels respectively correspond to environments with full daylight, overcast daylight, indoor working or retail environments, indoor transitory environments (e.g. hallways), and pitch black environments. We confirmed the illuminance of each condition with the light meter as described in the next section, and the measured values were found to vary less than 1% away from the intended illuminance value.

Displayed Color (14 levels): The displayed color on the OST-HMD was varied between 14 different colors, including black (appears as transparent [23]), white, the primary colors (red, green, blue), the secondary colors (yellow, cyan, magenta), and the tertiary colors (orange, chartreuse, spring green, azure, violet, and rose). All primary, secondary, and tertiary colors were presented at maximum saturation and value.

Together, these measurements also allow us to evaluate the performance of the model presented in the proceeding sections, where predicted contrast values can be compared against contrasts calculated from these measurements.

4.3. Procedure

First, for each targeted environment lighting condition, the light meter was positioned to be 8 centimeters in front of the light source, and the light source was dimmed until the illuminance measurement was representative of the targeted illuminance value. Five illuminance measurements were then recorded of the light source for that environment lighting condition.

Second, the HoloLens 2 was positioned as described in section 4.1, and the color to be presented on the display was configured via Unity. A series of five sequential illuminance measurements were then recorded. Following this, the displayed color was changed to the next condition and the process was repeated.

Once all colors had been measured for a particular environment lighting condition, the light source was dimmed according to the next condition and the process was repeated until the measurements for each condition were recorded.

4.4. Results and Discussion

In this section, we present the data collected from the photometric measurements as illuminance values and as Michelson contrasts..

Table 1. This table shows the average *illuminance* values, in lx, for the environment, black, white, primary colors, secondary colors, and tertiary colors respectively on the HoloLens 2 across lighting conditions ranging from 0 to 10,000 lx. Color names are abbreviated to their first letter for the primary colors (red, green, and blue), as well as for the secondary colors (yellow, cyan, and magenta). The tertiary colors are abbreviated to two letters representing the primary and secondary colors they fall between in the HSV color space. These tertiary colors are respectively: orange (R-Y), chartreuse (Y-G), spring green (G-C), azure (C-B), violet (B-M), and rose (M-R).

	Black/White		Primary Colors			Secondary Colors			Tertiary Colors					
Env.	Blk	W	R	G	B	Y	C	M	R-Y	Y-G	G-C	C-B	B-M	M-R
0	0	174.9	34.9	96.1	31.3	137.0	134.0	65.7	57.7	110.6	106.0	49.0	36.2	36.8
100	27.0	222.5	61.3	130.5	60.5	182.1	164.6	96.1	92.1	142.3	131.6	84.3	68.6	70.2
500	125.5	275.6	148.7	203.0	154.0	221.7	217.0	166.8	153.0	177.4	176.9	157.0	147.0	141.2
1,000	286.0	473.0	322.9	372.9	318.7	394.0	399.7	324.7	304.6	357.2	352.5	320.5	307.4	307.6
10,000	2665	2720	2625	2778	2747	2761	2743	2687	2651	2667	2663	2677	2674	2637

Table 2. This table similarly shows the *Michelson contrast* values between each color and the black level of the display on the HoloLens 2.

	White	Primary Colors			Secondary Colors			Tertiary Colors					
Env.	W	R	G	B	Y	C	M	R-Y	Y-G	G-C	C-B	B-M	M-R
0	0.997	0.985	0.994	0.983	0.996	0.996	0.992	0.991	0.995	0.995	0.989	0.985	0.986
100	0.783	0.388	0.657	0.383	0.742	0.718	0.561	0.546	0.681	0.659	0.515	0.435	0.444
500	0.374	0.085	0.236	0.102	0.277	0.267	0.142	0.099	0.172	0.170	0.112	0.079	0.059
1,000	0.246	0.061	0.132	0.054	0.159	0.166	0.063	0.031	0.111	0.104	0.057	0.036	0.036
10,000	0.010	0.008	0.021	0.015	0.018	0.014	0.004	0.003	<0.001	<0.001	0.002	0.002	0.005

4.4.1. Illuminance Measurements

Table 1 and figure 3 depict the mean (M) illuminance measurements made on the HoloLens 2 according to the environment illuminance and displayed color. The results show how the illuminance from the user's perspective changes with respect to the color of the light being emitted by the display and the intensity of the environment lighting. Due to illuminance measurements being weighed according to human color sensitivity, green-hued colors have higher illuminance values than either red or blue hued colors. This decrease in sensitivity is represented in the figure as the troughs that appear between white and yellow (for red), and between cyan and magenta (for blue). We see that the brightest illuminance is achieved at the display's white point, where all three color channels are set to maximum intensity, followed by the secondary colors yellow and cyan, where two color channels are set to maximum intensity, and then green-hued tertiary colors, such as chartreuse and spring green, where the green color channel is set to maximum intensity while another is set to half intensity. The minimum illuminance is always when displaying black, in which no light is emitted by the display, and is thus representative of the attenuated environment light as seen by the observer (kI_{env}, as represented in equation 9).

In general, this pattern in perceived intensity between the different displayed colors, which is particularly apparent for the 0 lx environment lighting plot, is inherent to display technology. This pattern arises due to the manner in which humans are sensitive to colors of different wavelengths [37]. We are most sensitive to green light compared to red and blue, and this sensitivity is inherently factored in for illuminance measurements, therefore we see increased illuminance for green-hued colors compared to red or blue hued colors. A change in the maximum illuminance that can be achieved by the display would have the effect of stretching or shrinking the vertical dimension of this plot, meaning that different displays will have slightly different plots, however the relative differences between colors should remain relatively stable, as long as similar primary colors are used to create the color spaces of the displays.

A subset of these tested conditions combine to form the term **a**, as seen in equation 9, for the HoloLens 2. These are the measurements of the illuminance contributed from the red, green, and blue colors at the 0 lx environment lighting level. As shown in section 3, these particular measurements represent the amount of light contributed solely by the display and can be used to model the illuminance for any displayed color when used in a manner such as the one shown in equation 7.

The term I_{env}, as represented in equation 9, is also measured several times in the procedure and forms the left-most column in table 1. These represent direct measurements of illuminance from the observer's perspective in the physical environment

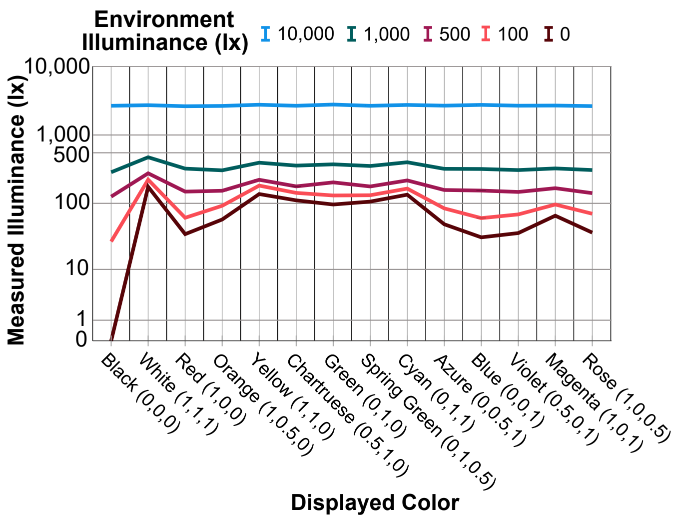


Fig. 3. This figure depicts the mean illuminance values for each of the 14 tested colors at each of the 5 environment lighting conditions. The *y*-axis is in log base 10 scale, with one additional label added to depict 500 lux.

prior to any attenuation performed by the display (i.e., without looking through the OST-HMD).

Additionally, the term k, as represented in equation 9, can be calculated based on the ratio of the measurements made of the display rendering black/transparent to the I_{env} measurement for that particular lighting condition ($k = I_{black}/I_{env}$). Based on the average of these ratios, gathered from columns 1 and 2 of table 1, we calculated that $k = 0.268$ for the HoloLens 2. As described in section 3, this term describes how the illuminance of the environment light is reduced after passing through the visor of the OST display.

4.4.2. Luminance Contrast

Table 2 and figure 5 (dashed lines) depict the Michelson contrasts calculated using equation 1 from the illuminance measurements presented in the previous section. These contrasts compare the illuminance from the user's perspective with the device powered on and rendering black (transparent) to the illuminance from the user's perspective with the display rendering white or one of the primary, secondary, or tertiary colors. In this manner, the term I_{max} corresponds to the sum of the illuminance of the light emitted by the display and the attenuated environment light, whereas I_{min} corresponds to solely the illuminance of the attenuated environment light, since no additional illuminance is added by the display when rendering black. These contrast values shown in table 2 are broken down according to the environment lighting level and displayed color. These particular contrasts compare a point in the virtual image to a point in the observer's view of their physical environment (see physical-virtual contrast in figure 1), and are thus representative of how easily distinguishable virtual imagery shown in the displayed color will be compared to the observer's view of their physical environment through the display.

As expected, contrast reduces in the presence of higher environment illuminance. While color choice has little impact on contrast in dark environment lighting conditions, it greatly affects contrast for lighting conditions between 100 and 1000 lx. Finally, for 10,000 lx environment conditions, we again see little variation in contrast with regard to the displayed color.

Note that additional contrasts could be similarly calculated between any pair of colors represented in table 1, as opposed to comparing each color to black (transparent). However, for the sake of space, these contrasts are not represented in this paper. These contrasts would be slightly different, in that they compare between two points in the virtual image (see virtual contrast in figure 1), as opposed to between a point in the virtual image and a point in the user's physical environment (see physical-virtual contrast in figure 1). These contrasts are therefore representative of how easily distinguishable virtual features shown in one color are with respect to the color of the virtual imagery surrounding the feature. For instance, how easily distinguishable a red symbol is when displayed over a uniform blue virtual background.

5. Analyzing Model Predictions

In this section we present an analysis of the accuracy of our model by comparing its predicted contrasts to the contrasts calculated in the previous section and presented in table 2.

5.1. Parameter Selection

In section 4.4.1, we demonstrated how photometric measurements can be made in order to establish values for the **a**, I_{env}, and k terms of equation 9. For our analysis, we used values of $\mathbf{a} = [35, 96, 31]$, based on the average illuminance measurements of the primary colors in the 0 lx environment condition (table 2), $I_{env} \in \{0, 100, 500, 1000, 10000\}$ environment lighting conditions, and $k = 0.268$ based on the average of the ratios of I_{black} to I_{env} values as calculated in section 4.4.1.

However, to proceed with the analysis we still need to specify the display's gamma correction function G, as seen in equation 7 and within the terms $\mathbf{x}$ and $\mathbf{y}$ of equation 9. We were not able to locate any information on the specific gamma correction function employed by the HoloLens 2. However, several standards exist for color and image encoding, the most widely used is the standard red, green, blue (sRGB), which provides standards for web-based imagery [35]. sRGB uses a gamma correction function in the form of $G(x) = ((x + c_1)/c_2)^{\gamma}$, where $c_1 = 0.055$, $c_2 = 1.055$, and $\gamma = 2.4$. Similarly, the BT.709 standards, which specify display standards for HDTVs, use a gamma correction function of the same form where $c_1 = 0.099$, $c_2 = 1.099$, and $\gamma = (1/0.45)$ [36].

To evaluate whether either of these functions resembled the one employed by the HoloLens 2, we performed several additional illuminance measurements in the same manner as those from the previous section for each of the primary colors at the 0 lx environment lighting condition at RGB values of 0, 0.25, 0.5, 0.75, and 1. We then normalized the data from the three color channels to be bound between 0 and 1 by dividing them by the maximum illuminance of their respective color channel (illuminance when R,G, or B equals 1). The normalized data for the three color channels was then averaged together to create a single plot representing the normalized gamma-corrected illuminance of the display, which is shown in green in figure 4. Figure 4 also shows the plots of the gamma correction function specified by the sRGB standards in red, and the BT.709 standard in blue. R squared and residual standard errors were calculated between the measured data and each of the other functions, where it was found that the measured data fits the sRGB standard function with $RSE = 3.8\%$ and $R^2 = 99.4\%$, and the measured data fits the BT.709 standard function with $RSE = 5.2\%$ and $R^2 = 98.9\%$.

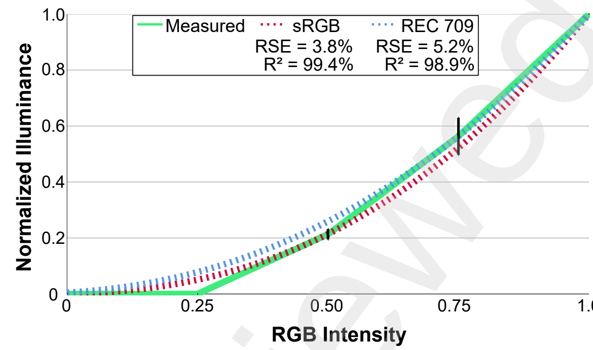


Fig. 4. This figure depicts a comparison between the measured gamma function of the HoloLens 2 (green solid plot), the sRGB standard gamma function (red dashed plot), and the BT.709 standard gamma correction function (blue dashed plot). Error bars (black) represent standard error with a multiplier of 1. Residual standard error and R squared error are reported.

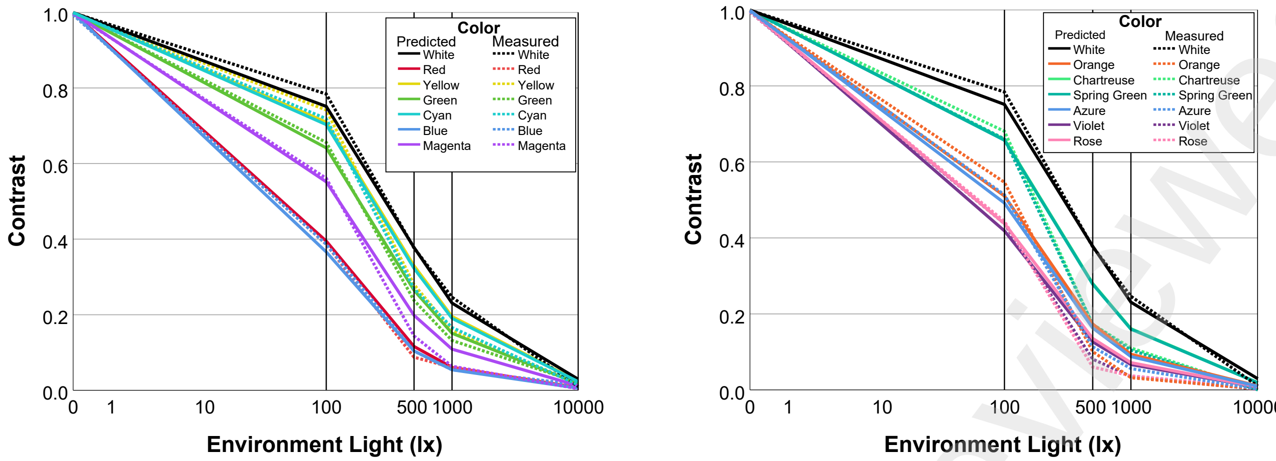
Since the sRGB correction function proved to be a better fit to our measured data, we decided to proceed with the analysis using it to define our gamma correction function G.

5.2. Analysis Results and Discussion

Graphs comparing the model's predictions to the contrast values established in section 4.4 can be seen in figure 5, separated into two graphs, one consisting of the primary and secondary colors while the other shows the tertiary colors. The results of our analysis indicated that the model using the parameters specified in the previous section achieved a residual standard error of 3.19% and $R^2 = 98.0\%$.

The model accurately predicts contrasts for the parameters specified according to the photometric measurements made on the HoloLens 2 and the assumptions made about its gamma correction function. However, it is important to consider possible sources of error, including the measurement accuracy of the illuminance meter used in section 4, and the gamma correction function used to predict the illuminance of different colored virtual imagery from the observer's perspective.

With regard to the illuminance meter, the manufacturer reports that the device has an accuracy of ±3% of the measured value. We performed repeated measurements and used averaged data to help eliminate portions of this error that arise due to random sampling error. Any systematic error could be problematic if we were presenting raw illuminance data, but since contrasts make comparisons between illuminance data, such a



(a) Primary and Secondary Colors (b) Tertiary Colors

Fig. 5. A comparison of the model's predicted contrasts (solid lines) to the measured contrasts (dashed lines) described in section 4.4. Each graph shows contrasts that compare each of the colors in the legend to the black level of the display at environment illuminance values of 0, 100, 500, 1,000, and 10,000 lx. The x-axis is shown in logarithmic scale for increased separation between our data points.

systematic error should not affect the accuracy of the predicted contrasts.

With regard to the gamma correction function, we chose a function specified by the sRGB standards that many traditional displays adhere to. This function appears to fit the data that was measured on the HoloLens 2 with reasonable accuracy. However, it is possible that the HoloLens 2, and other OST displays, do not adhere to these standards, as OST displays are a relatively new class of display that have many unique characteristics compared to traditional CRT, LCD, or LED-based displays. Since the gamma correction function employed by the display is not reported by the manufacturer, we are left to model it ourselves via photometric measurements in a manner as discussed in section 5.1 and presented in figure 4. In this paper, we are aided by the assumption that the function is in the form specified by the sRGB standards, however without this assumption, we would be left to use methods such as non-linear regression to approximate the function. Such methods are prone to many intricate considerations, such as providing a kernel function and reasonable range for its parameters, and these decisions could significantly impact the accuracy of the luminance contrast model, since it relies on accurate prediction of illuminance for specified RGB values.

Our intention is to provide a model of luminance contrast for OST displays that is easy to use, and although making photometric measurements on a display is relatively easy, it still poses a barrier to entry for most people who would be interested in using the model. As mentioned in section 3, the model can be used with photometric terms of either luminance or illuminance. Although luminance specifications are commonly published by display manufacturers, the luminance of the individual color channels are not typically reported. Such values can be approximated through assumptions about the color space and chromaticity of the primary colors of the display, but these are different for each OST display. In order to provide more convenient access to the parameters associated with individual displays of interest, we will continue to and encourage others to publish any photometric measurements made on OST displays for public access.

6. General Discussion

As demonstrated in the previous sections, illuminance measurements can be made to gather the values of I_{rgb}, and I_{env}. However, for the I_{env} parameter, direct measurement may not even be necessary a majority of the time, since standards exist [3] for indoor lighting levels and outdoor lighting levels [4] so that typical environment illuminance values can be looked up

[3] https://www.gsa.gov/node/82715
[4] https://www.engineeringtoolbox.com/light-level-rooms-d_708.html

via a table. If such tables are used to calculate contrast values for the range of environments that the particular application on the OST display will be used in, it can be easily verified as to whether or not the colors within the virtual imagery the observer is looking at offer acceptable contrast levels for the user's context (i.e. symbol recognition, reading).

Accessibility guidelines made by the W3C recommend a minimum simple contrast ratio of 4.5 to 1 for reading web-based virtual content, which when converted to Michelson contrasts refer to values of 64%, therefore we recommend using this value as a starting point for a general guideline for making UI color design decisions for applications on OST HMDs. However, we emphasize that this is only a starting point and that future work should specifically evaluate contrast standards for various tasks on OST displays, since there are many factors that influence user perception of virtual imagery that do not necessarily apply for more traditional displays [17].

For practical use cases, the brightest of sunny outdoor lighting conditions (near 130,000 lx) represents the maximum environment illuminance likely to be observed on an OST AR display [38]. Without significant attenuation, most displays are likely going to have a much smaller range of environment lighting conditions in which the contrast of the displayed imagery is at acceptable levels. The model presented in this paper can be used to calculate the upper bound of this range by specifying a desired contrast, for example the 64% recommended by W3C, and setting the input **x** and **y** RGB values to the maximum range using values of (1,1,1) for white and (0,0,0) for black. Then the equation can be rearranged to solve for I_{env} to find the maximum environment lighting that the particular display can be used in while meeting the desired contrast value:

$$I_{env} = \frac{|\mathbf{a}(\mathbf{x}-\mathbf{y})^T| - C\mathbf{a}(\mathbf{x}+\mathbf{y})^T}{2kC} \tag{10}$$

For example, using the parameters for the HoloLens 2 outlined in section 5.1, and the above-mentioned values of **x**, **y**, and C, we can calculate the I_{env} parameter to be approximately 170 lx. This implies that in environments with more than 170 lx of illuminance from the observer's perspective, the contrast between the virtual imagery and the environment does not meet the desired contrast defined by the C parameter, which could lead to the observer having difficulty distinguishing virtual imagery from their surroundings. In scenarios with illuminance values greater than this, then there is nothing else that can be done at the software level to improve contrast to the desired level, and the user will either have to adjust the lighting in their environment (if possible), move to an environment with more suitable lighting conditions, or increase the attenuation factor of the display such as by applying a neutral density filter to reduce the incoming environment lighting. Note that when using equation 10, it is possible to get negative values for I_{env} when the $C\mathbf{a}(\mathbf{x}+\mathbf{y})^T$ term is greater than the $|\mathbf{a}(\mathbf{x}-\mathbf{y})^T|$ term. When this occurs, it means that there is not an environment lighting condition that can create the desired contrast C, for the display with parameters **a** and k, displaying colors **x** and **y**.

The model can also be used to evaluate the effects of manipulating the attenuation factor k, in different environment lighting conditions. For example, adding an additional neutral density filter to the display scales the k term by a factor $k' \in \mathbb{R}, 0 < k' < 1$. Since applying an additional filter reduces the contrast of the user's view of their physical environment (see physical contrast in figure 1), the above equation can be rearranged to find the minimum k' for a desired contrast level, striking an optimal balance between reduced contrast in the user's physical environment and improved contrast in the virtual imagery:

$$k' = \frac{|\mathbf{a}(\mathbf{x}-\mathbf{y})^T| - C\mathbf{a}(\mathbf{x}+\mathbf{y})^T}{2kI_{env}C} \tag{11}$$

Similarly to equation 10, it is possible to get negative values of k' when the second term in the numerator is greater than the first. In such a case, this implies that even in a pitch black room, the two colors being compared, **x** and **y**, will not achieve the desired contrast level C. Additionally, positive values of k' that are greater than 1 imply that the desired contrast is achievable with the factored in attenuation factor k, and no additional filter k' is needed to achieve the desired contrast.

6.1. Design Guidelines and Considerations

Based on the understanding provided by our model, we have several guidelines for designing UIs and other types of virtual imagery to be viewed on OST displays.

In general, the contrast of any virtual element compared to the observer's view of their environment will be maximized when using combinations of black/transparent and white virtual imagery. As demonstrated above, this combination will result in the widest range of usable environment lighting conditions. For the specific case of UIs, it would seem intuitive that using either black features on a white background or white features on a black background would result in similar user performance and usability of the UI. However, Erickson et al. [23] demonstrated that "dark mode" style UIs with white features on black backgrounds offer several advantages in the form of increased visual acuity, increased usability, and increased visual comfort compared to light mode style UIs with white backgrounds and black features. Therefore, in general we recommend using dark mode style UIs with white features on black backgrounds.

When considering multiple colors, such as for a two-colored UI with a foreground/text color and a background color (see the tile-based UI design in figure 1), this pairing of colors should ideally be compared against a desired contrast level using equation 10 to find out if the contrast is acceptable for the desired environment lighting conditions, as well as find the max environment lighting condition that still offers acceptable viewing conditions. Should environment lighting exceed this maximum value, there are several options for further improving the contrast of the imagery:

Change Foreground to Green Hues: As green hues offer the highest illuminance second to white, changing the visual features in the foreground of imagery from red or blue tinted hues to green hues (between yellow and cyan in HSV color space) will offer improved contrast.

Decrease Saturation and Increase Value of Foreground: Alternatively, decreasing the saturation of the color while increasing the value of foreground visual features within the HSV color space has the effect of shifting the color towards white, thus increasing the contrast between the foreground and background of the virtual imagery.

Decrease Value of Background: Similarly, the background of the virtual imagery can be decreased in value within the HSV color space. Decreasing the value of the color has the effect of shifting the background color towards black, which increases the contrast between the background and foreground of the imagery. This also has the effect of reducing the opacity of the background.

If applying these techniques to the UI shown in figure 1, we would expect to see several changes to the UI's appearance. In general, the foreground colors used to represent text and most symbology in this screenshot are already displayed in white, however the appearance of the blue-tinted symbols, such as the ones on the leftmost column, could be either green-shifted and/or white-shifted to improve the contrast between the symbol and background. Further improvements to contrast could come about through manipulation of the background colors, where the color of the main tile and smaller tiles within could be shifted from blue towards black in order to improve the contrast between the items within the frame and the user's physical environment, although this could potentially interfere with the user's ability use Gestalt principles to group information in the way the designer intended them to [39].

Ideally, future OST displays will monitor environment lighting conditions through integrated sensors, and employ adaptive rendering techniques in order to optimize the appearance of virtual imagery for the user's current context. Such techniques could employ the above-mentioned strategies in order to make UIs more usable and legible for the user's context. However, there are other trade-offs to consider when recoloring imagery in this manner that should be investigated in future work. For example, it may not be advisable to adapt the appearance of color-encoded visual information, as shifting to higher contrast colors can potentially reduce the perceived difference between the colors in the imagery, which may result in users mistaking certain coded colors for others. There may be temporal concerns when adapting image color in this manner as well, as for example users may experience difficulty reading text that is dynamically changing colors. Additionally, dynamically changing the appearance of virtual imagery may introduce attentional cues that draw the attention of the user away from their current focus.

Similar methods could potentially be applied to other types of virtual imagery to ensure that visual features within them are easily distinguishable, but again there are trade-offs to consider when doing so that should be evaluated in future work. For example, uniformly changing the coloration of an image to consist of higher luminance colors increases the contrast between the image and the user's physical environment, but reduces the contrast within the image itself. Additionally, changing the coloration of 3D models and images to improve contrast could unintentionally affect the user's perception of this imagery in other more subtle ways, such as changing the user's attitude towards, or opinion of, the imagery, which may be important to consider for imagery that represents other people such as virtual avatars and video conferencing applications.

7. Conclusion

In this paper, we presented a model of luminance contrast that can be used to inform design decisions in the creation of virtual imagery as well as inform adaptive rendering techniques specific to OST displays. We presented the methods for determining the input parameters to the model through photometric measurements, and a set of measures for one consumer display, the Microsoft HoloLens 2. We hope that the results and model presented above will be valuable to future research as a convenient reference and tool for estimating the expected contrast observed by users of OST HMDs.

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Title: Modeling Luminance Contrast for Optical See-Through Displays
Submitted to: Computers and Graphics

Abstract:
Due to the additive light model employed by current optical see-through head-mounted displays (OST-HMDs), the perceived contrast of displayed imagery is reduced with increased environment luminance, often to the point where it becomes difficult for the user to quickly and accurately distinguish visual features within displayed virtual imagery. This problem is particularly important when it comes to user interface (UI) design, where the effectiveness of the UI is impacted both by the colors of the UI, the particular display being used, and the characteristics of the observer's physical environment. While existing contrast models can be used to predict user performance involving visual tasks on more traditional displays, they must be adapted for use with additive displays in order to take into account the blending of environment light with light emitted by the display. In this paper, we present a simplified model of luminance contrast for optical see-through displays that is derived from Michelson's contrast equation, and validate its performance by comparing predicted contrasts to contrasts calculated based on a set of photometric measurements made on a consumer OST-HMD, the Microsoft HoloLens 2. We demonstrate how this model can be used to inform UI design decisions by helping to select colors that are more robust to the observer's physical environment conditions, and demonstrate several additional use cases.

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